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Modified Transition State Theory of Reaction Rates*

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The specific reaction rate for chemical reactions is formulated in terms of the *total* transmission coefficient in the reaction coordinate and a modified transition state, which differs from the original transition state only in the treatment of the reaction coordinate. If the total transmission coefficient is factored into two partial transmission coefficients, the modified theory gives the transition state theory in both the classical and the quantum-mechanical-particle-in-a-box treatments. It is then shown that the momentum density in the virtual state over a symmetrical square barrier is that of a quantum-mechanical-particle-in-a-box at low momentum and that of a classical system at high momentum. Finally, the reaction rate transmission coefficient is calculated from the modified theory for an unsymmetric square barrier and shown to depend on the temperature, the activation energy, and the thermicity of the reaction.

I. DEVELOPMENT OF THE MODIFIED TRANSITION STATE THEORY

A CHEMICAL reaction may be represented by a flow of points (systems) over a potential energy hypersurface. On this surface are inscribed two hyper-areas, one of which describes the reactants and the other the products. The rate of the reaction is essentially the product of a transmission coefficient between the two areas and a frequency factor, i.e., the number of systems which strike the reactant boundary in unit time. Both of these quantities are complicated functions of all momenta, quantized and unquantized, as well as of position coordinates.¹

The multidimensional problem is reduced to a one-dimensional problem in the usual way by assuming that the reaction coordinate, X , and its conjugate momentum, p , can be separated from all other degrees of freedom.² It will then be assumed that there exists an intermediate called the "modified transition state" which is in equilibrium with the reactants in all degrees of freedom except the reaction coordinate. In this coordinate the intermediate is assumed to be in the reactant state; i.e., it moves as a free particle over a mean free path of length L at the energy of the reactant state. This distinguishes the modified transition state from the original transition state. The activation of the other degrees of freedom does not affect the activation energy since these are assumed to be in equilibrium with the reactants.

The number of systems which react per second is given by

$$R = \int \gamma(p_R) d\nu(p_R), \quad (1)$$

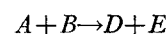
where $\gamma(p_R)$ is the total transmission coefficient and $d\nu(p_R)$ the number of systems with momentum p_R

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¹ Eyring, Walter, and Kimball, *Quantum Chemistry* (John Wiley and Sons, Inc., New York, 1944).

² For a discussion of the separation of coordinates, see E. Wigner, *Trans. Faraday Soc.* 34, 29 (1938).

approaching the reactant boundary in unit time.[†] For the reaction



it is given by

$$d\nu(p_R) = [AB\Gamma] \times [\exp(-p_R^2/2mkT) dp_R L/h] [p_R/Lm]. \quad (2)$$

The first bracket gives the number of systems in the modified transition state. $\Gamma = Q_c^\pm/Q_A Q_B$ is the ratio of the partition function for the modified transition state without the reaction coordinate to that for the reactant state. The second bracket is proportional to the probability that a system in the modified reactant state will have momentum p_R , and the last bracket gives the frequency for traversing the mean free path L .

Since

$$R = k_r AB \quad (3)$$

$$k_r = (\Gamma/mh) \int_0^\infty \gamma(p_R) \exp(-p_R^2/2mkT) p_R dp_R. \quad (4)$$

Equation (4) may be written in terms of the momentum p at the highest point E_R on the reaction coordinate. See Fig. 1. Now

$$p_R^2 = p^2 + 2mE_R$$

and

$$p_R dp_R = p dp.$$

If leakage through the barrier is neglected

$$\gamma(p_R) = 0 \quad \text{for } p_R < (2mE_R)^{1/2}$$

so that

$$k_r = (\Gamma/mh) \exp(-E_R/kT) \times \int_0^\infty \gamma(p) \exp(-p^2/2mkT) p dp, \quad (5)$$

where $\gamma(p) = \gamma(p_R)$. Equation (5) is the basic equation in the modified transition state theory.

[†] The statement "with momentum p_R " is to be taken to mean with momentum lying between p_R and $p_R + dp_R$.

II. DEVELOPMENT OF THE TRANSITION STATE THEORY

The transition state theory may be obtained from the modified theory as follows: The total transmission coefficient $\gamma(p)$ is broken down into the product of two partial transmission coefficients

$$\gamma(p) = \rho(p)\kappa(p). \quad (6)$$

The second of these $\kappa(p)$ may be taken to be the usual reaction rate transmission coefficient, the ratio of the number of systems which become products to the number which reach the top of the barrier. $\rho(p)$ is the ratio of the number of systems which reach the top of the barrier to the number which approach the barrier. It may be regarded as the density of momentum levels in the transition state normalized to the reactant state. Substitution of (6) in (5) and multiplication of the numerator and denominator by the arbitrary constant dq gives

$$k_r = \Gamma \exp(E_R/kT)/(mdq) \times \int_0^\infty \rho(p)\kappa(p) \exp(-p^2/2mkt) p dp dq/h. \quad (7)$$

This is the basic equation of the original transition treatment. It is customarily obtained as follows:

$$R = \int \kappa(p) dv'(p), \quad (8)$$

where

$$dv'(p) = [AB\Gamma \exp(-E_R/kT) \times [\rho(p) \exp(p^2/2mkt) dp dq/h] [p/mdq], \quad (9)$$

the number of particles per second, with momentum p in the forward direction and lying in the length dq of the transition state. The factors have the same meaning as in Eq. (2) except that reference is made to the transition state rather than to the modified transition state.

Integration of Eq. (8) together with Eq. (3) yields Eq. (7). Equation (7) may also be written

$$k_r = \langle [\kappa(p)p]_{\text{av}} \Gamma \exp(-E_R/kT)/mdq \rangle Q_t(p)/2, \quad (10)$$

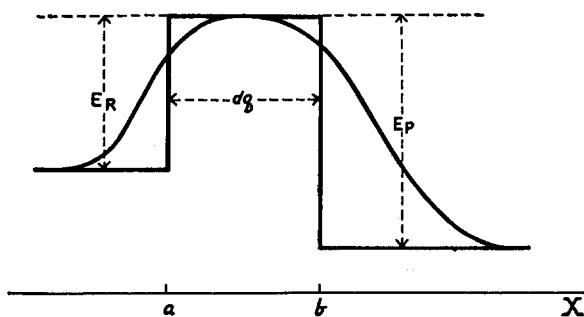


FIG. 1. Reaction coordinate section through the potential energy surface.

where

$$Q_t(p) = 2 \int_0^\infty \rho(p) \exp(-p^2/2mkt) p dp dq/h, \quad (11)$$

the translational partition function for the transition state and the bar denotes the average defined by the partition function.[‡]

In the early treatments, the following definition was made

$$\int \kappa(p)\rho(p) \exp(-p^2/2mkt) p dp dq/h = \kappa_A \int \rho(p) \exp(-p^2/2mkt) p dp dq/h. \quad (12)$$

κ_A may be called the mean transmission coefficient. Two quite different momentum density functions have been employed:

A. The classical momentum density:

$$\rho(p) = 1. \quad (13)$$

Substitution in (7) or (10) yields directly

$$k_r = \kappa_A \Gamma \exp(-E_R/kT) kT/h. \quad (14)$$

This is the generalization of the results of Peltzer and Wigner⁴ as given in Fowler.⁵

B. The momentum density for the quantum-mechanical-particle-in-a-box:

$$\rho(p) = (1/2)\delta(p - nh/2dq), \quad (15)$$

where δ is the Dirac delta function, which has the following properties.⁶

$$\delta(p - nh/2dq) = 0 \quad \text{unless} \quad p = nh/2dq \quad (16)$$

$$\int_{p_{n-1/2}}^{p_{n+1/2}} \delta(p - nh/2dq) p dp dq/h = 1 \quad (17)$$

$$\int_{p_{n-1/2}}^{p_{n+1/2}} f(p) \delta(p - nh/2dq) p dp dq/h = f(nh/2dq). \quad (18)$$

The factor (1/2) in front of the delta function corrects the density from that of the quantum-mechanical-particle-in-a-box, for which the momentum may take both signs, to the density of the transition state for which the momentum takes only one sign.[§]

³ It has been customary to associate the factor (1/2) with the average momentum rather than the partition function. See Glasstone, Laidler, and Eyring, *Theory of Rate Processes* (McGraw-Hill Book Company, Inc., New York, 1951).

[‡] This follows from the definition of the average

$$\bar{X} = [\int \{ \exp(-p^2/2mkt) \} X(p) dp] / Q.$$

⁴ H. Peltzer and E. Wigner, *Z. physik. Chem.* **B15**, 445 (1932).

⁵ R. H. Fowler, *Statistical Mechanics* (Macmillan Company, New York, 1936), second edition.

⁶ E. C. Kemble, *Fundamental Principles of Quantum Mechanics* (McGraw-Hill Book Company, Inc., New York, 1937).

[§] In the classical density the factor of (1/2) is introduced by integration over momentum from zero to infinity instead of from minus infinity to plus infinity.

Substitution of (15) in Eqs. (7) or (10) yields directly Eq. (14). This is the development used by Eyring in his original work.⁷

III. THE TRANSITION STATE AS A VIRTUAL STATE

In the previous section it was shown that a transition state in the reaction coordinate is obtained in the modified theory by breaking up the total transmission coefficients into two partial transmission coefficients. One of these defines the density of momentum levels in the transition state. It is of interest to calculate this function for several potential surfaces and compare the results with the density functions A and B which were assumed by earlier workers.

In transition state theory, the transition state is localized in a length dq of constant height E_R with the thermicity unspecified. These minimum specifications are met by the symmetrical square barrier. The effect of modifying the barrier will be discussed later.

The determination of the wave functions for this surface is straightforward.^{8,9} The wave functions in the three regions of Fig. 1 are

$$\begin{aligned}\psi_R &= A \exp(ip_R X/\hbar) + B \exp(-ip_R X/\hbar), \\ \psi_T &= C \exp(ip_X/\hbar) + D \exp(-ip_X/\hbar), \\ \psi_P &= E \exp(ip_P X/\hbar).\end{aligned}\quad (19)$$

At the boundaries the wave functions must be continuous, so that at a , $\psi_R = \psi_T$ and $d\psi_R/dX = d\psi_T/dX$ while at b , $\psi_T = \psi_P$ and $d\psi_T/dX = d\psi_P/dX$. These four boundary conditions give four relations among the five coefficients. The density of momentum levels, that is, the ratio of the number of systems on the top of the barrier to the number in the incident beam, is given by

$$\rho(p) = \frac{(|C|^2 + |D|^2)p}{|A|^2 p_R}. \quad (20)$$

For the symmetric square barrier $p_R = p_P$, and

$$\rho(p) = \frac{\frac{1}{2}(p/p_R + p_R/p)}{1 + (1/4)(p/p_R - p_R/p)^2 \sin^2(pdq/\hbar)}. \quad (21)$$

This momentum density has the following interesting properties: (1) For large values of the momentum it approaches the classical value of one. (2) At low values of the momentum it approaches the $(1/2)\delta(p - nh/2dq)$ of the quantum-mechanical-particle-in-a-box. This is borne out from an inspection of Fig. 2 and the fact that||

$$\int_{p_{n-1/2}}^{p_{n+1/2}} \rho(p) dp dq / h = \frac{1}{2}. \quad (22)$$

⁷ H. Eyring, J. Chem. Phys. 3, 107 (1937).

⁸ D. Bohm, *Quantum Theory* (Prentice-Hall Inc., New York, 1951).

⁹ J. O. Hirschfelder and E. P. Wigner, J. Chem. Phys. 7, 616 (1939).

|| For a finite dq it is not necessary to consider any variation except that of the sine term in carrying out the integration.

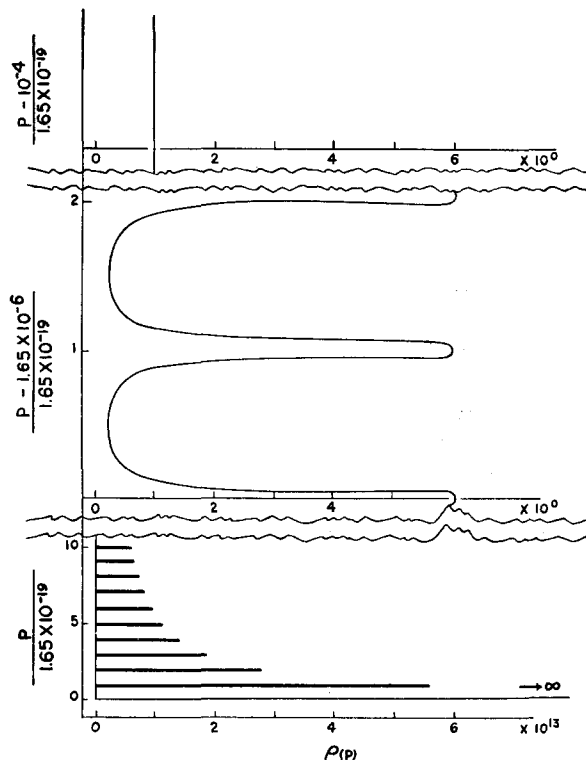


Fig. 2. The momentum density as a function of momentum over a symmetric square barrier.

(3) When p is averaged over one oscillation||

$$\bar{\rho}(p) = \frac{p_{n+1/2}}{p_{n-1/2}} \int \rho(p) dp / (h/2dq) = 1 \quad (23)$$

the result required by both the classical and particle-in-a-box treatments. It follows then that $\bar{\kappa}(p) = \bar{\gamma}(p)$.¶

(4) The energy half-width of the bands is roughly equal to the uncertainty in the energy which one calculates from the lifetime τ of a system in the transition state⁽¹¹⁾.

$$\Delta E \cong p \Delta p / m \cong p_R p^2 h / \pi m dq (p_R + p)^2 \quad (24)$$

$$\Delta E = h / \tau \cong h / (m dq / p T) \cong 4 p_R p^2 h / m dq (p_R + p)^2. \quad (25)$$

It is concluded that the virtual state with a momentum level density obtained from a symmetric square barrier gives a satisfactory description of the transition state.

For the unsymmetrical square barrier⁹

$$\rho(p) = 2 p p_R (p^2 + p_P^2) / [p^2 (p_R^2 + p_P^2) \cos^2(pdq/\hbar) + (p^2 + p_R p_P)^2 \sin^2(pdq/\hbar)]. \quad (26)$$

Here the similarity to the classical and to the quantum-mechanical-particle-in-a-box is much less marked. While $\rho(p)$ has peaks at $p = nh/2dq$ as for the symmetrical barrier, it has in addition smaller peaks at $p = (n + \frac{1}{2})h/2dq$. Further, neither of the integrals (22 or

¶ It should be noted that $\bar{\kappa}(p)$ is not equal to T the transmission coefficient for a single edge.

23) gives the values demanded by transition state theory. For a rounded barrier there is also little similarity.

The similarity between the virtual state for a square barrier at low momenta and the quantum-mechanical-particle-in-a-box deserves some comment. In both cases the energy levels arise from constructive and destructive interference from waves reflected at the boundaries. In the box no particles can exist except at the discrete levels, whereas in the virtual state, particles possessing energy other than the discrete values are reflected away from the top of the barrier. This selective reflection is fairly common, occurring in interferometers, the scattering of electrons by gases (Ramsauer effect), many nuclear reactions, x-ray scattering from solids, and in the theory of metals.

IV. THE CALCULATION OF THE SPECIFIC REACTION RATE FOR THE UNSYMMETRIC SQUARE BARRIER

The specific reaction rate for the unsymmetric square barrier is calculated from the modified transition state theory by equation

$$k_r = (\Gamma/mh) \exp(-E_R/kT) \times \int_0^\infty \gamma(p) \exp(-p^2/2mkT) p dp. \quad (5)$$

For this barrier

$$\gamma(p) = 1/[\dot{p}^2(p_R^2 + p_P^2) \cos^2(pdq/\hbar) + (p^2 + p_R p_P)^2 \sin^2(pdq/\hbar)]. \quad (27)$$

Since the other factors in the integrand vary slowly with p , it is permissible to replace $\gamma(p)$ by its average

over one oscillation,

$$\bar{\gamma}(p) = \frac{p_{n+\frac{1}{2}}}{p_{n-\frac{1}{2}}} \int \gamma(p) dp (h/2dq) = 4p p_R p_P / [p^2 + p_R p_P (p_R + p_P)]. \quad (28)$$

For low momentum and high barrier $\bar{\gamma}(p)$ may be expanded in terms of $p^2/(E_R E_P)^{\frac{1}{2}}$. Equation (5) then yields

$$k_r = \Gamma \exp(-E_R kT) (kT/h) \kappa_A, \quad (29)$$

where

$$\kappa_A = \frac{(4kT)^{\frac{1}{2}}}{(E_R)^{\frac{1}{2}} + (E_P)^{\frac{1}{2}}} \left[1 - \frac{9}{4} \left(\frac{kT}{E_R E_P} \right) + \frac{75}{32} \left(\frac{13}{5} + \frac{E_R + E_P}{(E_R E_P)^{\frac{1}{2}}} \right) \left(\frac{kT}{(E_R E_P)^{\frac{1}{2}}} \right)^2 - \dots \right]. \quad (30)$$

There are several points of interest in these results. The first is the kT/h factors out, just as in the Eyring theory. The factor kT/h has less significance here, however, and it is perhaps just as significant to treat the integral divided by mh as the frequency unit. The mean transmission coefficient κ_A is shown for a square barrier to have in general a value less than one, to increase with temperature and to decrease with increasing activation energy E_R and exothermicity, $E_R - E_P$. Rounding the barrier increases κ_A .⁹ Further, preliminary calculations appear to indicate that rounding the barrier decreases the effect of temperature, activation energy and thermicity on the mean transmission coefficient.

It should perhaps be emphasized that the conclusions drawn here are only as valid as is a one-dimensional reaction rate theory.